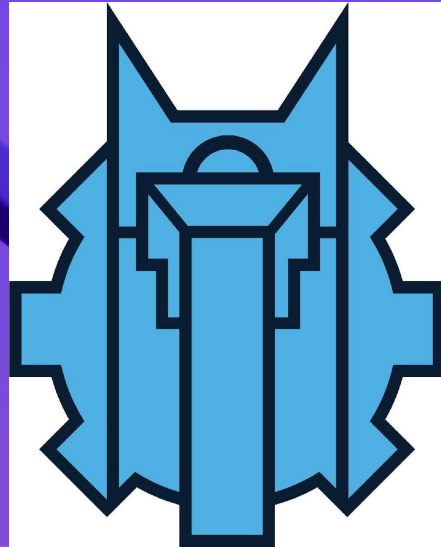


DevOps, 12-Factor, and Platform Engineering



Nashville
**DevOps
Days** 2024



Justin Reock

Head of DevRel

Cortex.io

Justin has over 20 years of experience working in various software roles including JEE work. He is an outspoken free software evangelist, delivering enterprise solutions, technical leadership, various publications and community education on databases, architectures, and integration projects.



SCAN ME

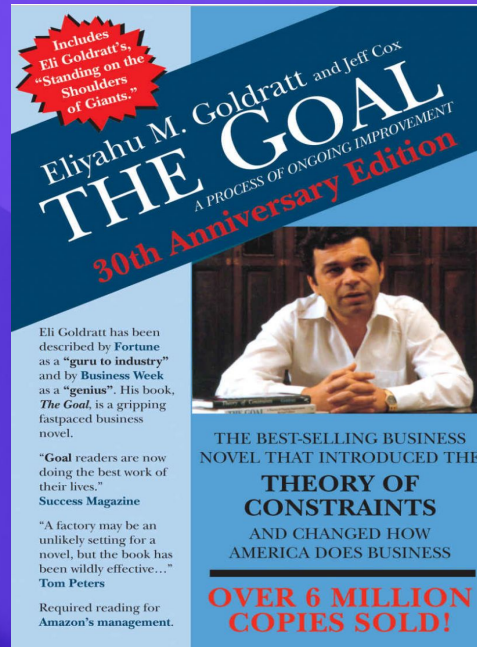




**“It’s no longer the big beating the small,
but the fast beating the slow.”**

Eric Pearson, CIO, InterContinental Hotels Group

The Ancient Business Wisdom ...Of the 70s and 80s



A Novel About IT,
DevOps, and Helping
Your Business Win

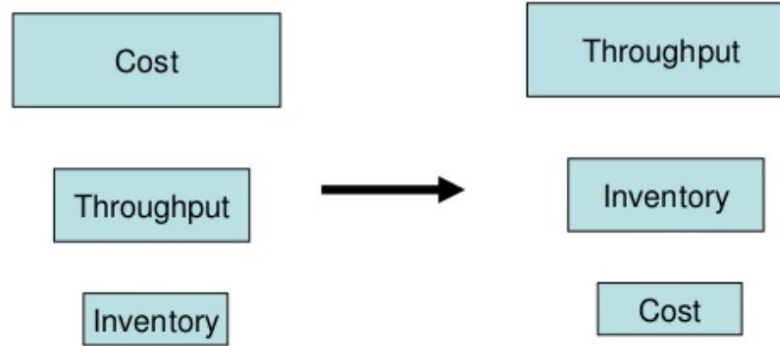
The Phoenix Project

Gene Kim, Kevin Behr,
and George Spafford



5TH
ANNIVERSARY
LIMITED
EDITION

Change in Focus
from the “cost world” to the “throughput world”

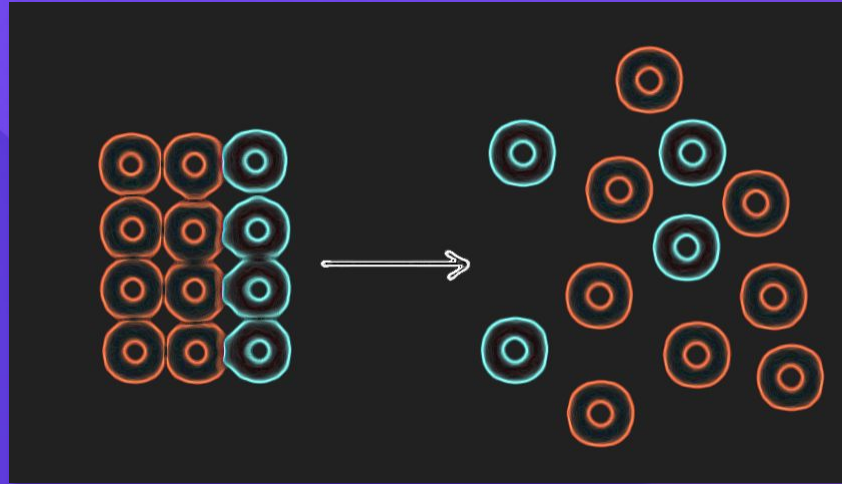


Cost = Organizational Cost

Inventory = Code

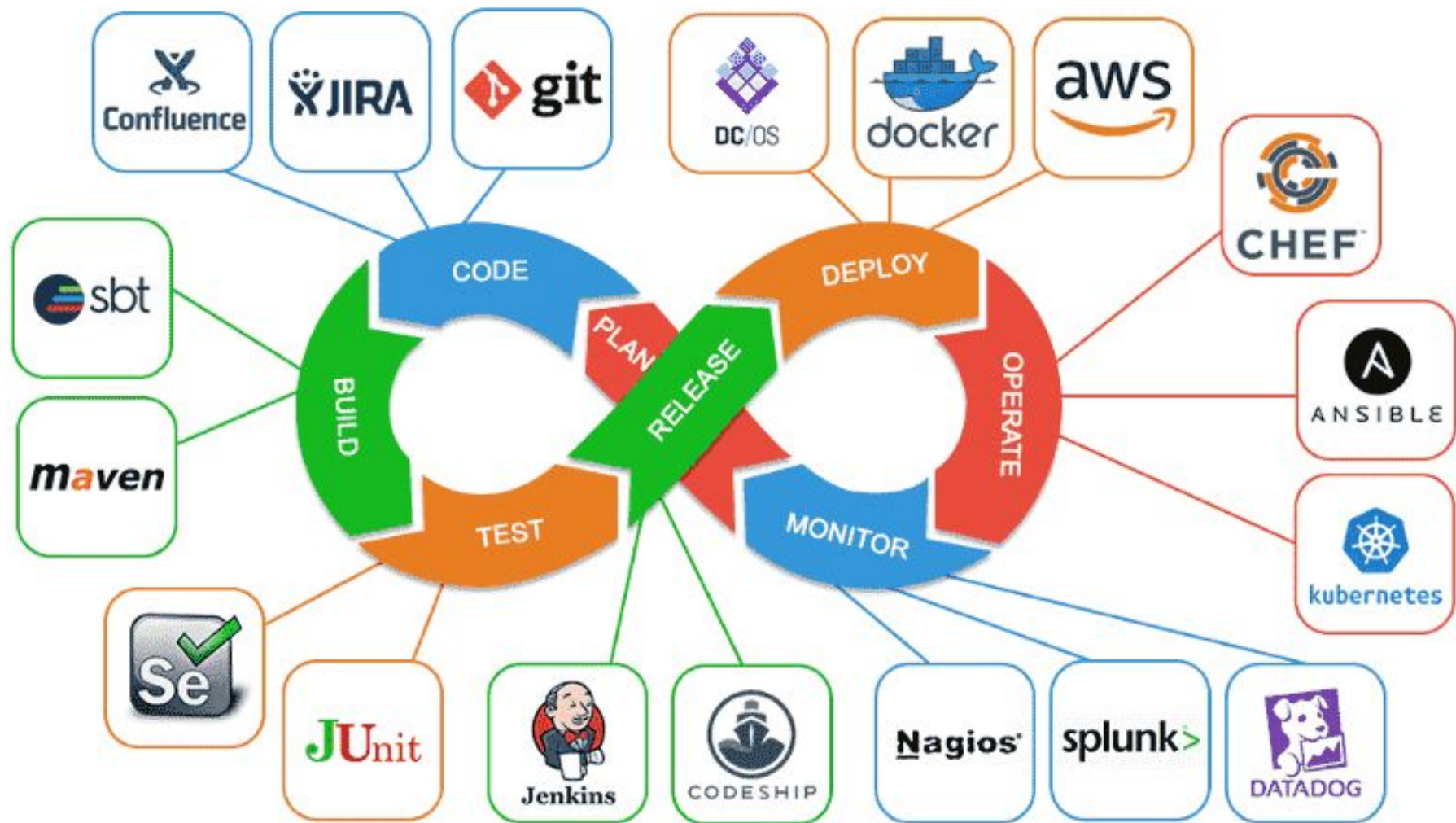
Throughput = Money

It doesn't really matter what you improve as long as you are constantly improving something, because...



...entropy denotes that if you aren't doing anything, you're always getting worse. Feedback is key to improvement...

The only initiatives that will positively impact long-term performance are ones which increase throughput while simultaneously decreasing cost.



CNCF Cloud Native Interactive Landscape

The cloud native landscape ([png](#), [pdf](#)), serverless landscape ([png](#), [pdf](#)), and member landscape ([png](#), [pdf](#)) are dynamically generated below. Please [open a pull request](#) to correct any issues. Greyed logos are not open source. Last Updated: 2022-11-07T16:

You are viewing 1,179 cards with a total of 3,415,658 stars, market cap of \$18.9T and funding of \$54.4B.

Landscape

Card Mode

Members

Serverless

Wasm

Database

Streaming & Messaging

Application Definition & Image Build

Continuous Integration & Delivery

App Definition and Development


CNCF Graduated


CNCF Graduated


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Ultra-Portable: How has this changed software?



THE TWELVE-FACTOR APP

INTRODUCTION

In the modern era, software is commonly delivered as a service: called *web apps*, or *software-as-a-service*. The twelve-factor app is a methodology for building software-as-a-service apps that:

- Use **declarative** formats for setup automation, to minimize time and cost for new developers joining the project;
- Have a **clean contract** with the underlying operating system, offering **maximum portability** between execution environments;
- Are suitable for **deployment** on modern **cloud platforms**, obviating the need for servers and systems administration;
- **Minimize divergence** between development and production, enabling **continuous deployment** for maximum agility;
- And can **scale up** without significant changes to tooling, architecture, or development practices.

The twelve-factor methodology can be applied to apps written in any programming language, and which use any combination of backing services (database, queue, memory cache, etc).

The 12 Factors Lead to Portable Design

I. Codebase

One codebase tracked in revision control, many deploys

II. Dependencies

Explicitly declare and isolate dependencies

III. Config

Store config in the environment

IV. Backing services

Treat backing services as attached resources

V. Build, release, run

Strictly separate build and run stages

VI. Processes

Execute the app as one or more stateless processes

VII. Port binding

Export services via port binding

VIII. Concurrency

Scale out via the process model

IX. Disposability

Maximize robustness with fast startup and graceful shutdown

X. Dev/prod parity

Keep development, staging, and production as similar as possible

XI. Logs

Treat logs as event streams

XII. Admin processes

Run admin/management tasks as one-off processes

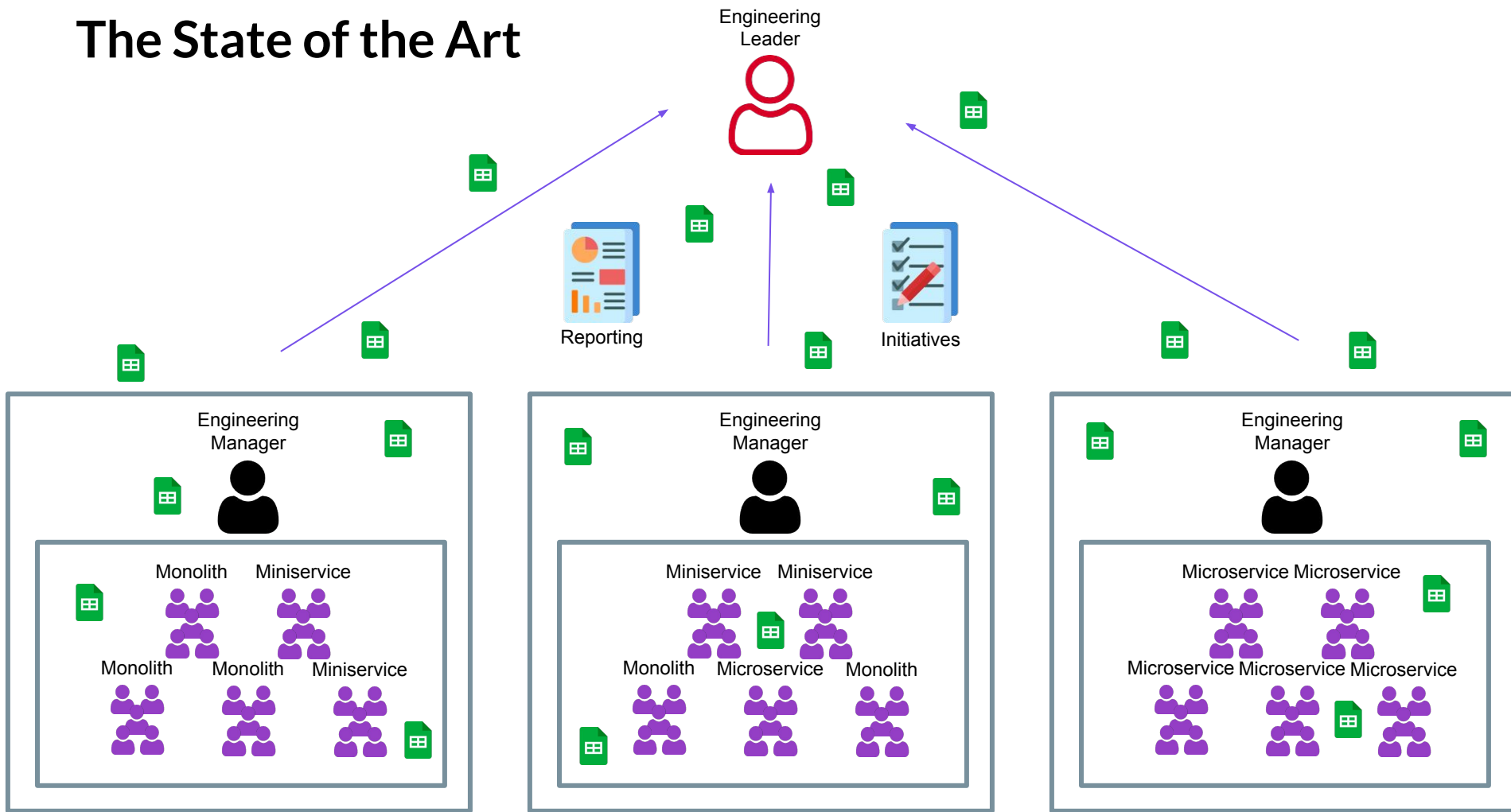


Freedom and Portability

Freedom and Portability
+
Complexity and Disorganization


$$\begin{aligned} &\text{Freedom and Portability} \\ &+ \\ &\text{Complexity and Disorganization} \\ &= \\ &\text{Toil, Friction, and Context} \\ &\quad \text{Switching} \end{aligned}$$

The State of the Art





What if we had a single interface that controlled,
monitored, and operationalized every service in
your platform?

cortex.io



cortex